

Def: Given two polynomials $p(x)$ and $q(x)$ with degrees d_p and d_q respectively, and if $p(x) \neq q(x)$, then number of points where they intersect is less than or equal to $\max(d_p, d_q)$

NOTE: Schwartz Zippel lemma holds for polynomials in finite fields.

Use Case: ① Checking polynomial equality in $O(1)$ time. (Probabilistic)
↓
probability of choosing a point where two polynomials are same is very less in a finite field if $d \ll p$

② Vector equality.
(Lagrange interpolation +)
pt. 1

